



PAMANTASAN NG LUNGSOD NG SAN PABLO
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NUMERICAL METHODS

CPE 312

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Lecturer



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FALSE-POSITION METHOD



Learning Objectives

- *Define False-Position Method
- *Determine and perform the algorithm of False-Position Method
- *Solve functions using False-Position Method



- Regula-Falsi Method
- Oldest method (dates back to ancient Egyptians)
- Also known as method of linear interpolation



- Effective alternate to Bisection method.
- Roots convergence is faster than Bisection method.
- Computational efforts are less.



False-Position Method Algorithm

1. Find points x_0 and x_1 such that
 $x_0 < x_1$ and $f(x_0) \cdot f(x_1) < 0$.
2. Take the interval $[x_0, x_1]$ and find next value using

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$



3. If $f(x_2)=0$ then x_2 is an exact root,
else if $f(x_0) \cdot f(x_2) < 0$ then $[x_0, x_2]$,
else if $f(x_2) \cdot f(x_1) < 0$ then $[x_2, x_1]$.
 4. Repeat steps 2 & 3 until $f(x_i) \approx 0$.
- Stopping criterion $f(x_0)$ & $f(x_2) = 0.000\dots$



Example-1

Find a root of an equation

$f(x) = x^3 - x - 1$ using False Position method.



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Step 1.

$$f(x) = x^3 - x - 1$$



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$$f(0) = 0^3 - 0 - 1 = -1$$



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$$f(x) = x^3 - x - 1$$

$$f(0) = 0^3 - 0 - 1 = -1$$

$$f(1) = 1^3 - 1 - 1 = -1$$

$$f(2) = 2^3 - 2 - 1 = 5$$



Step 1.

$$f(x) = x^3 - x - 1$$

$$f(0) = 0^3 - 0 - 1 = -1$$

$$f(1) = 1^3 - 1 - 1 = -1$$

$$f(2) = 2^3 - 2 - 1 = 5$$



Step 1.

$$f(x) = x^3 - x - 1$$

$$f(0) = 0^3 - 0 - 1 = -1$$

$$f(1) = 1^3 - 1 - 1 = -1$$

$$f(2) = 2^3 - 2 - 1 = 5$$

Initial point **[1, 2]**

[illegible]

[illegible]



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$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$



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$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$
$$x_2 = \frac{(1)(5) - (2)(-1)}{5 - (-1)}$$



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$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$
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$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$
$$x_2 = \frac{(1)(5) - (2)(-1)}{5 - (-1)}$$
$$x_2 = \frac{5 - (-2)}{5 - (-1)}$$
$$x_2 = \frac{7}{6} = 1.16667$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]



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Iterations	x_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	1	-1	2	5	1.16667	-0.57869
2	1.16667	-0.57869	2	5	1.25311	-0.28537
3	1.25311	-0.28537	2	5	1.29344	-0.12953
4	1.29344	-0.12953	2	5	1.31128	-0.05659
5	1.31128	-0.05659	2	5		



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4	1.29344	-0.12953	2	5	1.31128	-0.05659
5	1.31128	-0.05659	2	5	1.31899	



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6	1.31899		2			



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4	1.29344	-0.12953	2	5	1.31128	-0.05659
5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	



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5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228		2			



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5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	



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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442



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5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368		2			



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7	1.32228	-0.01037	2	5	1.32368	-0.00442
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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	



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Iterations	x_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	1	-1	2	5	1.16667	-0.57869
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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187



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Iterations	x_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	1	-1	2	5	1.16667	-0.57869
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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428		2			



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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5		



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1	1	-1	2	5	1.16667	-0.57869
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6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5	1.32453	



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7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5	1.32453	-0.00080



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8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5	1.32453	-0.00080
10	1.32453		2			



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9	1.32428	-0.00187	2	5	1.32453	-0.00080
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9	1.32428	-0.00187	2	5	1.32453	-0.00080
10	1.32453	-0.00080	2	5	1.32464	



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2	1.16667	-0.57869	2	5	1.25311	-0.28537
3	1.25311	-0.28537	2	5	1.29344	-0.12953
4	1.29344	-0.12953	2	5	1.31128	-0.05659
5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5	1.32453	-0.00080
10	1.32453	-0.00080	2	5	1.32464	-0.00033



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Iterations	x_0	$f(x_0)$	x_1	$f(x_1)$	x_2	$f(x_2)$
1	1	-1	2	5	1.16667	-0.57869
2	1.16667	-0.57869	2	5	1.25311	-0.28537
3	1.25311	-0.28537	2	5	1.29344	-0.12953
4	1.29344	-0.12953	2	5	1.31128	-0.05659
5	1.31128	-0.05659	2	5	1.31899	-0.02430
6	1.31899	-0.02430	2	5	1.32228	-0.01037
7	1.32228	-0.01037	2	5	1.32368	-0.00442
8	1.32368	-0.00442	2	5	1.32428	-0.00187
9	1.32428	-0.00187	2	5	1.32453	-0.00080
10	1.32453	-0.00080	2	5	1.32464	-0.00033



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The approximate root of
 $f(x) = x^3 - x - 1$ is 1.32464.



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1. $f(x) = xe^x - 1$

2. $f(x) = x^3 - 2x^2 - 5$



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Reminder:

Submit your Activity # 6

<https://forms.gle/9W6kPvSw7wUeB9E87>

on or before 5:30 pm.



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